

Cyanoacrylate glue reactions: A systematic review, cases, and proposed mechanisms

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ABSTRACT

Objective: Cyanoacrylate glue closure was first used in humans 10 years ago to treat venous reflux of the axial veins. Studies have since shown its clinical efficacy in vein closure. However, great need exists to elucidate further the types of specific adverse reactions that cyanoacrylate glue can cause for better patient selection and to minimize these events. In the present study, we systematically reviewed the literature to identify the types of reported reactions. In addition, we explored the pathophysiology contributing to these reactions and proposed the mechanistic pathway with inclusion of actual cases.

Methods: We searched the literature for reports of reactions following cyanoacrylate glue use in patients with venous diseases between 2012 and 2022. The search was performed using MeSH (medical subject headings) terms. The terms included cyanoacrylate, venous insufficiency, chronic venous disorder, varicose veins, vein varicosities, venous ulcer, venous wound, CEAP (clinical, etiologic, anatomic, pathophysiologic), vein, adverse events, phlebitis, hypersensitivity, foreign body granuloma, giant cell, endovenous glue-induced thrombosis, and allergy. The search was limited to the literature reported in English. These studies were evaluated for the type of product used and the reactions noted. A systematic review, in accordance with the PRISMA (preferred reporting items for systematic reviews and meta-analyses) method, was performed. Covidence software (Melbourne, VC, Australia) was used for full-text screening and data extraction. Two reviewers reviewed the data, and the content expert served as the tiebreaker.

Results: We identified 102, of which, 37 reported on cyanoacrylate use other than in the context of chronic venous diseases and were excluded. Fifty-five reports were determined appropriate for data extraction. The adverse reactions to cyanoacrylate glue were phlebitis, hypersensitivity, foreign body granuloma, and endovenous glue-induced thrombosis.

Conclusions: Although cyanoacrylate glue closure for venous reflux is generally a safe and clinically effective treatment choice for patients with symptomatic chronic venous disease and axial reflux, some adverse events could be specific to the properties of the cyanoacrylate product. We propose mechanisms for how such reactions can occur based on histologic changes, published reports, and case examples; however, further exploration is necessary to confirm these theories. (*J Vasc Surg Venous Lymphat Disord* 2023;11:876-88.)

Keywords: Adverse events; Cyanoacrylate; Foreign body granuloma; Hypersensitivity; Phlebitis

Treatment methods for axial venous reflux continue to evolve to achieve effective outcomes less invasively with fewer adverse effects and shorter convalescence periods. The nonsurgical closure of axial veins can be broadly classified into two types: thermal tumescent and nonthermal, nontumescent (NTNT). The advantage of NTNT is the lack of tumescent anesthesia and no risk of thermal nerve injury. The newest form of NTNT is cyanoacrylate glue closure (CAC).

Cyanoacrylate adhesive was initially used for wound closure starting in the 1950s.^{1,2} In 2000, the first internal use was approved for application in arterial venous malformations as a liquid embolic system (TruFill; Cordis Neurovascular, Inc, Johnson & Johnson, Co, Miami Lakes, FL). The first case of truncal vein closure in a swine model was reported in 2011.³ A first-in-human feasibility study of 38 patients receiving great saphenous vein (GSV) closure with cyanoacrylate showed a 94.7% closure rate in

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3 years.⁴ Subsequently, a European postmarketing survey showed an 88.5% 3-year closure rate in 70 patients.⁵ The randomized control study comparing a cyanoacrylate glue device (VenaSeal Closure System; Medtronic) and a radiofrequency ablation device (ClosureFast; Medtronic) has demonstrated an early (3-month) closure rate of 99% and 5-year closure rate of 91.4% for VenaSeal.⁶⁻⁸

Three CAC products are currently on the market. VenaSeal (Medtronic) received the Conformité Européenne Mark in 2011 and Food and Drug Administration approval in 2015. VenaSeal became covered by the Centers for Medicare & Medicaid Services in 2018. As of April 2021, >250,000 procedures using VenaSeal have been performed worldwide in >80 countries. The other two products are VenaBlock and VariClose, which are from Turkey and not approved by the Food and Drug Administration.

Cyanoacrylate comes in four forms: methyl-, ethyl-butyl-, and octyl-cyanoacrylate. They are used industrially, commercially, and medically. The butyl- and octyl-cyanoacrylates are used in healthcare. Its unique property is that it remains liquid at room temperature but has the distinct characteristic of instantaneously transforming into a solid, strong adhesive on contact with moisture. This feature causes cyanoacrylate to have a liquid-like consistency in a monomer state. However, on interaction with moisture or negatively charged ions, this will polymerize and harden to become a solid adhesive material. Cyanoacrylate closure differs from other treatment modalities because it introduces a permanent material that behaves like a bioimplant once deployed. Thus, it can elicit a unique effect profile mediated by the immune system. With the increased use of this treatment modality, novel adverse effects are being identified. Our study aims to increase our understanding and awareness regarding these adverse effects, hypothesize the mechanisms, and consider the guidance that could affect the management of adverse events (AEs) and patient selection.

METHODS

We searched the MEDLINE, Cochrane, and SCOPUS databases for clinical studies, case series, and case reports in English that reported AEs and reactions following cyanoacrylate glue use in patients with venous diseases between 2012 and 2022. The search was performed using MeSH (medical subject headings) terms. The terms included cyanoacrylate, venous insufficiency, chronic venous disorder, varicose veins, vein varicosities, venous ulcer, venous wound, CEAP (clinical, etiologic, anatomic, pathophysiologic), vein, adverse events, AEs, phlebitis, hypersensitivity, foreign body granuloma, giant cell, endovenous glue-induced thrombosis (EGIT), allergy. Additionally, a manual search was performed, and the reference lists of the studies were reviewed to identify additional studies. These studies were evaluated for the

type of product used and the type of reactions noted. Covidence software (Melbourne, VC, Australia) was used for full-text screening and data extraction. Two reviewers reviewed the data, and the content expert served as the tiebreaker.

RESULTS

We identified 102 records, of which, 37 studies that had reported on cyanoacrylate use other than in the context of chronic venous diseases were excluded (Fig 1). We determined that 55 reports were appropriate for data extraction. Details of our literature review of the identified reports of the reactions after the cyanoacrylate procedure to treat superficial venous reflux are presented in the Table.

Phlebitis was the most commonly reported AE, with a wide variation in the reported incidence, which varied from 1.2% to 25%. However, given the paucity of a uniformly agreed-on definition, some investigators have classified phlebitis separately from thrombophlebitis, and others have not. Barring small studies, the incidence of EGIT and infection were generally low and comparable to the incidence noted after thermal ablation.^{55,59} Other reported AEs included hematomas, ecchymosis, pigmentation, pain, diffuse rash, asthma, atopic dermatitis, and granulomas.

DISCUSSION

From the events reported in the literature, the AEs related to cyanoacrylate glue can be divided into the following: phlebitis, hypersensitivity (types I and IV), foreign body granuloma, EGIT, and infection. We provide hypotheses for the mechanisms of how these reactions occur and suggest theories on why some patients will develop severe reactions but others will not, because such information could help guide better patient selection.

Phlebitis. Phlebitis is the most common AE after CAC, with a higher occurrence rate than after endothermal ablation techniques. The reported rate varies from 0.8% to 12% after endothermal ablation techniques to 1.2% to 25% after CAC.^{4-6,9,10,60-62} Phlebitis is thought to represent inflammation of the vein or veins; however, no consensus has been reached on the definition of phlebitis. It is usually diagnosed from the signs and symptoms. However, even among trained personnel, inconsistencies exist in the levels of agreement.^{63,64} Similarly, wide variations exist in the signs and symptoms used in the literature to diagnose and report phlebitis after treatment of venous insufficiency or endothermal ablation techniques.^{60,65}

Early studies assessing the outcomes after CAC have not provided the definition used to diagnose phlebitis.⁶ Proebstle et al⁵ defined phlebitis as “reddening of the overlying skin and pain on palpation.” The median onset

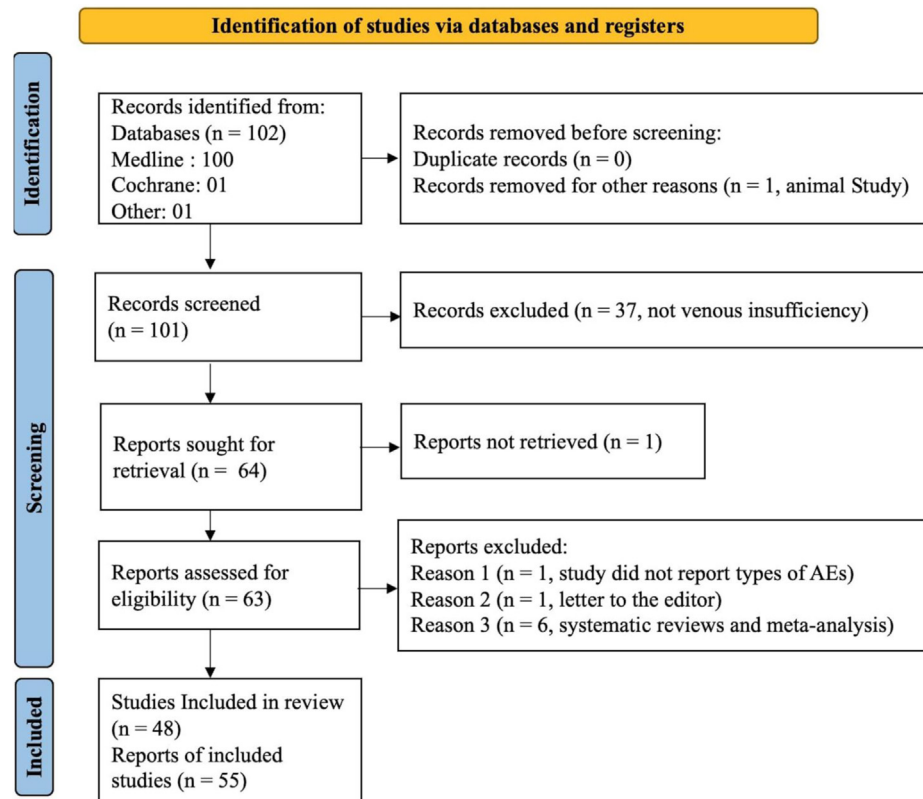


Fig 1. PRISMA (preferred reporting items for systematic reviews and meta-analyses) flow diagram.

of symptoms was 6 days after the procedure, and the reported median duration of symptoms was 6.5 days. However, it was gradually recognized that the dermal reaction after CAC was different from that identified as phlebitis after endovenous thermal ablation or sclerotherapy. Gibson and Ferris¹⁶ defined phlebitis as “pain, tenderness, and/or erythema in the treated segments or side branches.” They developed a phlebitis classification schema, where P1 was defined as phlebitis involving the truncal vein, P2 as phlebitis involving the tributaries and/or side branches of the truncal vein, and P3 as a nonspecific erythematous reaction.¹⁶ Park et al¹⁴ used the term “abnormal skin reaction” and phlebitis-like abnormal reaction (PLAR) to differentiate the dermal phenomenon noted after CAC. They defined it as erythema, itching, pain, edema, and pain and/or tenderness over the truncal vein area and hypothesized that it represented a foreign body or hypersensitivity reaction.¹⁴ Gibson et al³⁸ later defined a CAC hypersensitivity reaction as a “red, itchy dermal reaction that is sometimes painless but sometimes associated with discomfort and/or localized swelling.” Itching has been reported to be a prominent feature, occurring in 91% to 95% of patients with PLAR and/or a hypersensitivity reaction.^{31,49} Pruritus is a characteristic symptom of allergic or atopic diseases, although it is also noted in other settings.⁶⁶ Sermsathanasawadi et al⁴⁹ noted epifascial saphenous

veins with a subcutaneous distance between the anterior vein wall and the skin of <1 cm and saphenous vein diameter of ≥ 8 mm to be independent factors associated with a hypersensitivity reaction. Park et al³¹ found that the occurrence of PLAR was more frequent in cases with an epifascial GSV length >10 cm than in those with subfascial GSV and occurred more frequently in the GSV than in the superficial saphenous vein. Gibson et al³⁸ found a trend toward decreased hypersensitivity reactions in patients with CEAP clinical class C4 to C6. These variations have been thought to be related to possible differences in the definition and techniques used and racial distribution of those enrolled.

Based on the severity of the reaction as determined by the clinician, no treatment, over-the-counter medications, nonsteroidal anti-inflammatory drugs, antihistamines, topical steroids, oral steroids, and, in rare cases of recurrent phlebitis, vein excision have been used to treat this condition. Premedication with intravenous steroids did not reduce the incidence of PLAR in one study.³¹

Hypersensitivity. The ideal healing process after CAC is as follows. After a short acute inflammatory phase, chronic inflammation with dampening of the immune response will begin and, over time, lead to complete occlusion of the vessel with fibrous tissue and/or thrombus,

Table. Reported adverse events (AEs) after cyanoacrylate glue closure (CAC) of incompetent lower extremity veins

Investigator; study type	Product	Phlebitis, PLAR, HSR, thrombophlebitis, dermal reactions ^a	EGIT	Infection	Other reported AEs
Almeida et al, ⁴ 2013; OCS	VenaSeal	7/38 (18.4)	8/38 (21)	1 (2.6)	Hyperpigmentation, n = 1
Morrison et al, ⁶ 2015; RCT	VenaSeal	20/108 (18.5)	0	1/108 (<1)	Paresthesia, n = 3
Proebstle et al, ⁵ 2015; OCS	VenaSeal	8/70 (11.4)	1/70 (1.4)	1/70(1.4)	Pain, n = 5; bruising, n = 1
Bozkurt et al, ⁹ 2016; RCT	VariClose	7/154 (4.5)	0	0	Pigmentation, n = 2; ecchymosis, n = 22
Tekin et al, ¹⁰ 2016; CS	VariClose	2/62 (3.2)	0	0	Hematoma, n = 1
Morrison et al, ¹¹ 2017; RCT	VenaSeal	5/108 (4.62)	0	0	Asthma, n = 1; erythema and rash, n = 2
Koramaz et al, ¹² 2017; RR	VariClose	3/150 (2.1)	0	0	Pain, n = 7
Almeida et al, ¹³ 2017; CS	VenaSeal	6/38 (15.8)	DVT 1/38 (2.6) ^b	1/38, injection site infection (2.6)	Atopic dermatitis, n = 1; skin ulcer, n = 2
Park et al, ¹⁴ 2017; CS	VenaSeal	8/34 (23.5)	0	0	Pigmentation, n = 4
Premnath et al, ¹⁵ 2017; CS	Endocryl	0	3/124 (2.4)	NA	NA
Gibson et al, ¹⁶ 2017; CS	VenaSeal	10/50 (20)	1/50 (2)	0	0
Chan et al, ¹⁷ 2017; CS	VenaSeal	4/108 (3.7) ^c	2/108 (1.8)	NA	NA
Yasim et al, ¹⁸ 2017; CS	VariClose	0	0	0	0
Bademci et al, ¹⁹ 2018; CS	VariClose	2/50 (4)	0	0	Ecchymosis, n = 1
Eroglu et al, ²⁰ 2018; RCT	VariClose	11/175 (6.5)	0	0	Ecchymosis, n = 9
Gibson et al, ²¹ 2018; RCT	VenaSeal	0	0	0	Shin splint, n = 1; erythema, n = 1
Yavuz et al, ²² 2018; RR	VenaBlock	6/538 (1.2)	0	0	Ecchymosis, n = 5
Korkmaz et al, ²³ 2018; RR	NBCA	0	0	0	No AE reported
Tang et al, ²⁴ 2018; IMG	VenaSeal	1/1 (100)	NA	NA	NA
Novotny et al, ²⁵ 2019; CR	NBCA	0	0	0	FBR, n = 2; pulmonary embolism, n = 1
Fletes et al, ²⁶ 2019; RCT	VenaSeal	23/103 veins (22) ^c	0	0	NA
Bademci et al, ²⁷ 2019; RR	VariClose	4/75 (5.3)	0	0	Pigmentation, n = 2; ecchymosis, n = 2
Pillutla et al, ²⁸ 2019; RR	VenaSeal	0	7/54 (13)	0	0
Yang et al, ²⁹ 2019; RR	VenaSeal	4/106 (3.7)	1/106 (<1)	3/106 (2.8)	Paresthesia, n = 3; FBC, n = 3
Hwang et al, ³⁰ 2019; CS	VenaSeal	10/63 (16.7) ^c	0	1/60 (1.7) ^c	Hyperpigmentation, n = 8
Park et al, ³¹ 2019; CS	VenaSeal	69/271 (25.4) ^c	0	0	0
Tang et al, ³² 2019; CS	VenaSeal	10/93 (10.8) ^c	1/77 (1.2) ^c	0	0
Calik et al, ³³ 2019; CS	TGK	7/200 (3.5)	0	0	Pigmentation, n = 7; ecchymosis, n = 24; paresthesia, n = 6
Jones et al, ³⁴ 2019; CR	VenaSeal	1/1 (100)	0	0	0
Nasser et al, ³⁵ 2019; CR	VenaSeal	1/1 (100)	0	0	NA
Lew et al, ³⁶ 2019; CR	VenaSeal	0	0	0	FBC, n = 1
Morrison et al, ⁷ 2019; RCT	VenaSeal	1/108 (<1)	0	0	Scar, n = 1
Weaver et al, ³⁷ 2019; CS	VenaSeal	22/107 (21)	0	0	0
Gibson et al, ³⁸ 2020; CS and RR	VenaSeal	18/286 (6)	0	0	20/286 (7% patients; 6.4% limbs): pain, tenderness, and swelling without erythema or itching; phlebitis without evidence of HSR
Cho et al, ³⁹ 2020; RR	VenaSeal	0	11/191 (5.8)	0	0
Tang et al, ⁴⁰ 2020; CS	VenaSeal	27/156 (17.3) ^b	2/156 (1.3) ^b	0	Ecchymosis, n = 45 ^b

(Continued on next page)

Table. Continued.

Investigator; study type	Product	Phlebitis, PLAR, HSR, thrombophlebitis, dermal reactions ^a	EGIT	Infection	Other reported AEs
Sumarli et al, ⁴¹ 2020; CS	VenaSeal; VenaBlock	0	0	4/4 (100)	2/4 Glue casts and presumed FBC
Parsi et al, ⁴² 2020; CS	VenaBlock	0	0	0	FBR, n = 1
Chan et al, ⁴³ 2020; CS	VenaSeal	11/37 (29.7)	1/37 (2.7)	0	Bruising, n = 2
Fiengo et al, ⁴⁴ 2020; CR	VenaSeal	1/1 (100)	0	0	0
Langridge et al, ⁴⁵ 2020; CR	VenaSeal	0	0	0	FBC, n = 1
Morrison et al, ⁸ 2020; RCT	VenaSeal	0	0	0	0
Almeida et al, ⁴⁶ 2020; CR	VenaSeal	0	0	0	FBR, n = 1
Proebstle et al, ⁴⁷ 2021; OCS	VenaSeal	0	0	0	0
Sermasathanasawadi et al, ⁴⁸ 2021; RR	VenaSeal	0	0	0	FBC, n = 3
Sermasathanasawadi et al, ⁴⁹ 2021; RR	VenaSeal	16/101 (15.8)	0	0	0
Linn et al, ⁵⁰ 2021; CS	VenaBlock	3/29 (7.7)	0	5/29 (17.2)	Bruising, n = 5
Tang et al, ⁵¹ 2021; CS	VenaSeal	4/90 (4.4)	0	0	Bruising, n = 3
Chen et al, ⁵² 2021; CR	VenaSeal	0	0	1/1 (100)	FBC, n = 1
Wilczko et al, ⁵³ 2021; CS	VenaBlock	0	0	0	0
Joh et al, ⁵⁴ 2022; RCT	VenaSeal	3/63 (4.7)	2/63(3.1)	0	Numbness, n = 1; pigmentation, n = 1
O'Banion et al, ⁵⁵ 2022; RR	VenaSeal	0	5/396 (1.3)	0	0
Yigit et al, ⁵⁶ 2022; RR	VenaBlock	3/78 (3.8)	0	0	0
Bahi et al, ⁵⁷ 2022; RR	VenaSeal	5/235 (21)	3/235 (1.27)	6/235 (2.5)	Neuropraxia, n = 9
Yen et al, ⁵⁸ 2022; CR	VenaSeal	0	0	0	Pulmonary embolism, n = 1

CR, Case report; CS, case series; DVT, deep vein thrombosis; EGIT, endovenous glue-induced thrombosis; FBC, symptomatic foreign body granuloma; FBR, asymptomatic foreign body reaction; HSR, hypersensitivity reaction; IMG, image; NA, not applicable; NBGA, n-butyl-2-cyanoacrylate; OCS, observational cohort study; PLAR, phlebitis-like abnormal reaction; RCT, randomized controlled trial; RR, retrospective review; TCK, Turkish glue kit. Data presented as number/total (%), unless noted otherwise.

^aPhlebitis, thrombophlebitis, and PLAR/HSR events were reported together, unless the authors defined and identified these events separately.

^bReported as not therapy or device related.

^cStudy reported incidence of event per vein or extremity.

with near complete resolution of inflammation and preserved integrity of the vein wall. However, dysregulation of immune responses can result in various degrees of inflammation and symptoms after the procedure. Often, the processes underlying the different types of hypersensitivities occur concurrently and can be clinically indistinguishable.

Type I hypersensitivity. Type I hypersensitivity is an immediate immune reaction and involves immunoglobulin E (IgE)-mediated release of antibodies against the soluble antigen. This results in mast cell degranulation and the release of inflammatory mediators such as histamine. Type I hypersensitivity can present in a biphasic manner. The first stage, also known as the immediate stage, usually occurs rapidly (within minutes of exposure). The late phase can develop 4 to 12 hours after the

early phase reaction, and the duration is ~24 to 72 hours.⁶⁷ It can be challenging to differentiate an actual type 1 hypersensitivity reaction from acute inflammation or other types of hypersensitivity after CAC deployment.

Degranulation of mast cells on vein histologic examination was noted in a vein dissected 10 minutes after CAC exposure.⁴² However, mast cell degranulation with histamine release is part of an acute inflammatory reaction following placement of bioimplant material and might not necessarily represent a hypersensitivity reaction.⁶⁸

The development of rhinoconjunctivitis, urticaria, and an asthmatic reaction after occupational exposure to cyanoacrylate-containing glue has been reported in the literature.^{69,70} One patient was noted to have developed hives all over the body within 1 week of CAC, which responded to treatment with antihistamines and

steroids.¹⁶ These reactions are thought to be type I hypersensitivity reactions. A type I reaction is evaluated using a skin prick test or serum IgE test.⁷¹ However, it has been reported that it is not possible to perform prick tests or specific IgE measurements with cyanoacrylates owing to the physical properties of the compounds.⁶⁹ Thus far, no cases of anaphylaxis or evidence of type 2 or 3 hypersensitivity after CAC have been reported.

Type IV hypersensitivity. Type IV hypersensitivity is also known as a delayed-type hypersensitivity reaction and is mediated by antigen-specific effector T cells. This type of hypersensitivity lags 1 to 3 days from exposure to the response.

Allergic contact dermatitis is a type IV hypersensitivity reaction. Acrylates are a common contact allergen group and were named the Contact Allergen of the Year in 2012. The specific acrylate, isobornyl acrylate, was named the 2020 Contact Allergen of the Year. Cyanoacrylates are acrylate derivatives and have been shown to cause contact dermatitis. To date, people with hypersensitivity reactions to cyanoacrylates have not been shown to have cross-reactivity with other acrylates. However, it has been hypothesized that exposure to incompletely polymerized acrylates (which can be found in various personal and commercial use products) can induce broad cross-reactivity.⁶⁷ All currently available cyanoacrylate-based products used for treating chronic venous disease contain *n*-butyl cyanoacrylate (NBCA) and proprietary additives that determine its characteristics, such as viscosity, polymerization time, and so forth. Different products could contain various amounts of residual monomers, which could affect the product's propensity to cause sensitization. Active monomer particles can also be released from polymer structures that could act as haptens. The extent to which additives contribute to hypersensitivity or immune-mediated reactions is unknown.

Histopathologic evidence has supported the theory of cell-mediated immune response. In a case report, a patient, after CAC of a left GSV, underwent resection of the vein at postoperative day 200 because of recurrent leg pain, erythema, and edema.³⁴ Immunohistochemical stains of the tissue showed that most mononuclear cells were T lymphocytes. Most of these were of the T4 subset, suggesting a cell-mediated type IV hypersensitivity reaction. In another case report, a GSV segment treated with cyanoacrylate was excised 5.5 years after the procedure. It showed that CD3 lymphocytes were distributed widely throughout the treated segment, forming clusters in the adventitia and dispersed across the medial layer into the lumen. In contrast, only a few CD3 lymphocytes were seen in the control segment.⁴⁶

A study reported a 2.2% incident rate of allergic contact dermatitis in patients receiving NBCA-containing topical skin adhesives after foot and ankle surgeries.⁷² The risk factors for allergic contact dermatitis have been divided

into acquired and inherent. Reported acquired risk factors include inflammatory skin conditions such as atopic dermatitis, stasis dermatitis, irritant contact dermatitis, and multisensitization (a history of hypersensitivity to multiple antigens as seen on patch tests).⁷³ In contrast, inherent factors are genetic variances that result in various degrees of susceptibility. Several investigators have attributed "atypical phlebitis" or "phlebitis-like acute reactions" to type IV hypersensitivity.

Foreign body granuloma. A foreign body reaction with the presence of foreign body giant cells after CAC is a physiologic response seen in all patients who receive this treatment, as noted in our tabular pathology data ([Supplementary Table](#), online only). Some investigators included foreign body granuloma as a subtype or the result of type IV hypersensitivity.⁷⁴⁻⁷⁶ However, it has been demonstrated in murine models that the formation of foreign body giant cells is a part of the innate immune system and can occur without T lymphocyte cells.⁷⁷ In vivo, the foreign body reaction results from a complex interplay between innate and adaptive immune systems. Significant cross-talk occurs between T lymphocyte cells and macrophages. This interaction determines whether the cells produce factors that promote the persistence of inflammation or those that enable the dampening of the immune response and wound healing.^{78,79} Some patients with foreign body reactions and vessel wall destruction are asymptomatic.⁴² In contrast, others can develop granulomas, skin breakdown, and glue cast extrusion.³⁶ In some cases, the hypersensitivity reactions seem to occur shortly after the procedure with no recurrence. However, others can have a relapsing course requiring repeated courses of treatment.³⁴ Thus, it is essential to attempt to identify the risk factors and the mechanisms that cause this exaggerated response.

Endovenous glue-induced thrombosis. EGIT has been defined as the proximal migration of glue–thrombus complex from the GSV toward the saphenofemoral junction (SFJ) with various levels of encroachment or thrombus extension into the deep vein.^{28,39} The appearance of EGIT has been described as "thread-like." However, subtotal occlusion of deeper veins has also been reported.⁸⁰ It has been hypothesized that the mechanism is inadvertent proximal glue migration before complete polymerization, possibly aided by procedural compression.^{4,39}

The literature is ambiguous regarding the nomenclature, because later publications referred to these as EGIT rather than deep vein thrombosis. This hesitation in using deep vein thrombosis could have therapeutic and prognostic implications, and no guidelines have been established to distinguish cyanoacrylate glue from thrombus on ultrasound.¹⁷

Studies have graded EGIT similar to the Kabnick and Lawrence classification of endovenous heat-induced

thrombosis (EHIT).^{28,39,81} Almeida et al⁴ reported a high incidence of EGIT and recommended the deployment of glue 5 cm distally from the SFJ and an additional pull-back of 1 cm between the first and second trigger.

A recent retrospective study found no statistically significant difference between the incidence of EGIT vs that of EHIT.⁵⁵ The variation in the reported incidence of EGIT after VenaSeal, VenaBlock, and VariClose might be due to small sample sizes, differences in polymerization time and techniques, and different populations, with the latter two products used mostly in people of Middle Eastern or Asian ethnicity. EGIT has been reported to develop within days of the procedure and can resolve within weeks to months, although in some cases, it has been shown to persist for >1 year.²⁸ Some investigators have surmised that based on its difference in appearance compared with EHIT, EGIT might have a lower risk of thrombus progression and extension.³⁹

Risk factors of advanced age, GSV diameter, and clotting history were reported in small retrospective studies. However, these findings have yet to be replicated. Given the difference in mechanisms that lead to the formation of EGIT, the risk factors for the development of EHIT might not be extrapolated to EGIT. Two studies have reported pulmonary embolisms.^{25,58}

No consensus guidelines have been established for the treatment of EGIT. Minimal data are available regarding the thrombogenicity of cyanoacrylate glue. However, in contrast to endovenous thermal ablation, CAC introduces a foreign body that persists in the intravascular space, which could affect the risk of thrombus formation and propagation and could have implications regarding management.

Because of the difference in the mechanism of thrombus formation, persistence of the foreign body, and risk of the development of complications, it is possible that the same approach for EHIT should not be extrapolated to EGIT. For management, some have favored periodic monitoring, and others have used anticoagulation based on the extent of thrombus formation. Cho et al³⁹ have proposed a classification of EGIT based on the glue-occupied area in the deep vein similar to that used for EHIT. No studies have reported the effectiveness of postprocedural duplex ultrasound scans on the outcomes.

Infection. Despite the theoretical risk of hematogenous seeding of bioimplants and biofilm creation, the reported incidence of infection after CAC has been comparable to the reported incidence of infection after endovenous thermal ablation techniques and other surgical site infections.^{61,58,82} However, caution must be exercised because the risk factors for surgical site infections, such as diabetes mellitus, increasing age, decreased physical activity, and tobacco use, also have a high prevalence in patients with chronic venous

disease.^{83,84} Moreover, recurrent and persistent bacteremia in the absence of any other source has been reported in patients who have received cyanoacrylate treatment for bleeding gastric varices, suggesting the possibility of seeding and/or colonization of cyanoacrylate glue.⁸⁵ It can be challenging to distinguish a local infection from a foreign body reaction after CAC. To date, infections secondary to *Staphylococcus aureus* (n = 2), *Pseudomonas aeruginosa* (n = 2), *Enterobacter cloacae* (n = 1), and *Citrobacter koseri* have been mentioned in studies that reported the isolated organisms.^{41,52}

Theories for exaggerated immune system reaction.

Several models or theories have been proposed to explain and/or understand why the body develops an immune response. No one model can fully explain all the immune responses, and these mechanisms likely act in concert.

One theory is the self/non-self model, which assumes that the body mounts an immune response to entities it considers non-self.⁸⁶ Although this would explain the body mounting an immune response to a foreign biomaterial such as NBCA, it does not explain the inconsistency of why only some people mount an exaggerated immune response and a relapsing response.

Another theory is a triggered inflammatory response due to a secondary stimulus. It has been reported that an increased proinflammatory state in the body due to an unrelated stimulus can provoke or amplify the body's immune response to hitherto inert foreign bodies or foreign bodies with a stable foreign body reaction. It has been suggested that the change in the immune or inflammatory status of an individual with conditions such as systemic infections, autoimmune diseases, vaccinations, and neoplasms could have downstream effects that somehow trigger the development of foreign body granulomas in individuals with cosmetic dermal fillers and other biomaterials that had been inert for years.⁸⁷⁻⁸⁹ A similar mechanism might explain why some people develop robust immune responses weeks to months after the index procedure or a recurrence of hypersensitivity symptoms after periods of quiescence.

The "danger model" could be another explanation. This model proposes that from an evolutionary perspective, whether an entity and the associated damage results in "normal, quiet" cell death or "messy" cell death determines whether an immune response is elicited. "Messy death" generates signals perceived as danger signals by antigen-presenting cells, which subsequently results in the uptake of antigen and upregulation of costimulatory molecules on antigen-presenting cells. If recognition of an antigen by T cells (signal 1) occurs, together with costimulation (signal 2), it results in T-cell activation. If cells undergo "normal cellular death," danger is not perceived, and the absence of signal 2 results in the development of tolerance.⁹⁰ It is possible that in some cases (perhaps in

the context of a secondary proinflammatory state), tissue damage from cyanoacrylate glue is perceived as “messy death,” resulting in T-cell activation. Drug-induced hypersensitivity reactions, autoimmune conditions, and transplant rejection responses are being reconsidered in this model.^{90,91}

An exaggerated response to the material itself is a possibility. Acrylates are plastics; these monomers are potent sensitizers that can be polymerized using various processes, and these polymerized products are thought to remain inert. However, because different methods result in various degrees of a fully polymerized final product, some products can contain more residual monomers than will others, which can affect their potential to sensitize.⁶⁷

The hapten hypothesis and p-i hypothesis have been proposed for drug-related hypersensitivity reactions.⁹¹ According to the hapten hypothesis, a chemically reactive drug or drug metabolite can form covalent bonds with a protein to form a hapten–carrier complex, resulting in the formation of an antigen capable of eliciting a T-cell response. The “p-i” hypothesis suggests that some drugs can bind to immune receptors and stimulate T cells. The presence of haptens and contribution to the development of allergic contact dermatitis has been suggested in the case of Dermabond (2-octyl cyanoacrylate and plasticizers; Ethicon, Franklin Lakes, NJ).^{91,92} Although possible, it is unknown whether these mechanisms contribute to adaptive immune system activation in the case of NBCA. No data are available regarding the presence of impurities within product formulations or their role in hypersensitivity reactions.

It is also possible that, like in diseases such as tuberculosis, leprosy, and sarcoidosis, tissue injury following CAC is primarily due to exaggerated immune responses rather than damage due to the material itself. In these examples, the sustained release of antigens and continued activation of sensitized T cells can result in increased tissue damage.⁹³

Another theory is that increased cyanoacrylate material interaction with subcutaneous tissue could contribute to an exaggerated response. Per instructions provided by manufacturers of various NBCA products, compression is applied close to the recently introduced cyanoacrylate glue several times during the procedure. The pressure applied depends on the operator and is difficult to standardize. Pressure applied during the process could force the cyanoacrylate glue into smaller tributaries that might not accommodate the glue, leading to extravasation into the extravascular space. This could subsequently lead to sensitization and granuloma formation. This has also been suggested by Actor and Langride et al.⁴⁵

CASE STUDY 1 (FOREIGN BODY GRANULOMA)

An 82-year-old man presented with nonhealing ulcers in his right lower extremity. The patient had initially developed skin breakdown after bumping his right leg 3 years prior and subsequently had recurrent cellulitis and skin breakdown. He is a long-time smoker with peripheral artery disease in his right leg (ankle brachial index, 0.7). Duplex ultrasound revealed a long >6-second GSV reflux extending from the SFJ to the ankle. He initially had undergone angioplasty of his superficial femoral artery stenosis to improve lower extremity runoff. Given his peripheral artery disease history, we elected to treat the GSV reflux with CAC with VenaSeal to avoid excessive postprocedure compression. Following treatment of GSV reflux with cyanoacrylate glue, his wound healing improved, although he had another setback with cellulitis due to *Pseudomonas aeruginosa*. His wounds had healed ~2 months after GSV closure with continued wound care. Approximately 1 month following wound closure (3 months after CAC), the patient developed a nontender, elastic, firm lump in the proximal medial calf. Over the next few weeks, this developed into a golf ball size lump, with skin breakdown over this area and a location in the distal calf. He also developed a similar lump in the right groin. Bedside ultrasound showed a hypoechoic mass surrounding the previously treated GSV (Fig 2, A). At that point, we suspected a foreign body granuloma formation. We attempted treatment with intralesional steroids, to which he showed some improvement. However, he subsequently developed another skin breakdown in an area along the treated GSV. Given that local intralesional treatments were not sufficiently controlling his symptoms, we opted to treat him with systemic steroid therapy. He was given a long prednisone taper regimen that lasted for 6 weeks (60 mg, 50 mg, 40 mg, 30 mg, 20 mg, and 10 mg for 1 week each). After the systemic steroid treatment, his leg symptoms appeared to improve for a short period. However, shortly after the treatment, his condition had worsened, with increased swelling, erythema, and drainage from skin ulcer sites (Fig 2, B). Computed tomography showed subcutaneous masses in areas below the ulcers (Fig 2, C). Given that he showed no improvement with high-dose systemic steroid treatment, we opted to perform a resection of the affected areas, including the groin mass and below-knee GSV, with recurrent skin breakdown and granuloma formation. The resected specimens were analyzed both macroscopically and microscopically. On macroscopic examination, the groin mass consisted of two cystic structures with a well-circumscribed capsule. Inside this capsule was a thick creamy substance in one and a dark, bloody substance in the other, both with multiple pellet-like material inside (Fig 3, A).

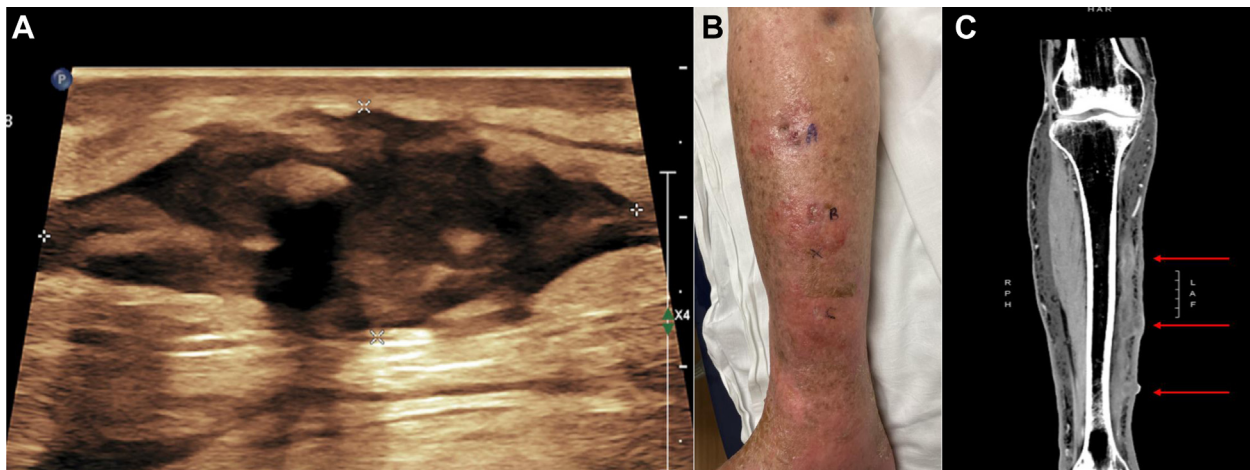


Fig 2. **A**, Ultrasound image of right medial knee mass showing a hypoechoic mass surrounding the cyanoacrylate-treated great saphenous vein (GSV). **B**, Photograph showing erythema, skin breakdown, and drainage developing in the right leg shortly after oral steroid treatment completion. **C**, Computed tomography scan of the right leg showing subcutaneous masses underneath areas of skin ulceration.

The resected groin mass and below-knee GSV section showed white to grayish chalky cottage cheese-like material (Fig 3, B). The tributary veins from the GSV connected to the area where the skin ulcerations had developed (Fig 4, A). Microscopic evaluation showed a large inflammatory collection around the vein.

Immediately deep into this reaction was a muscular vessel that contained foreign material associated with the foreign body giant cell reaction (Fig 4, B). This material was similar to that seen in a report of a resected saphenous vein after cyanoacrylate ablation.⁴⁶ Most of the material was in the lumen, but some fragments

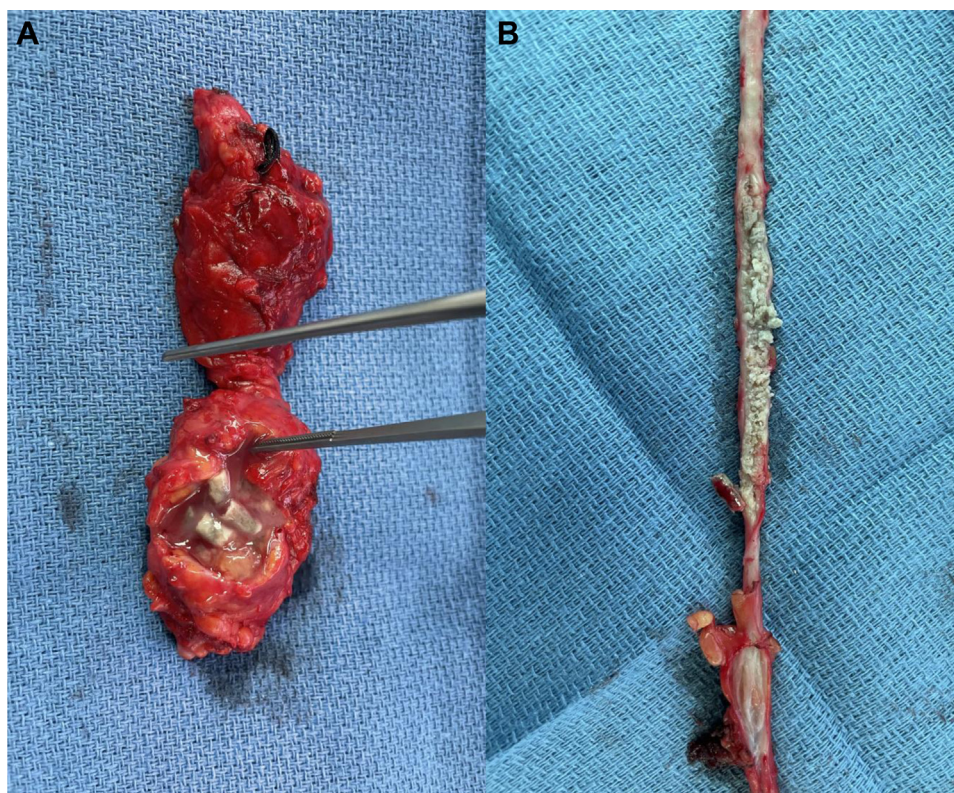


Fig 3. **A**, Resected groin mass, which presented as encapsulated tissue with white-gray pellet-like material inside. **B**, Below-knee right great saphenous vein (GSV) with white-gray chalk-like material inside.

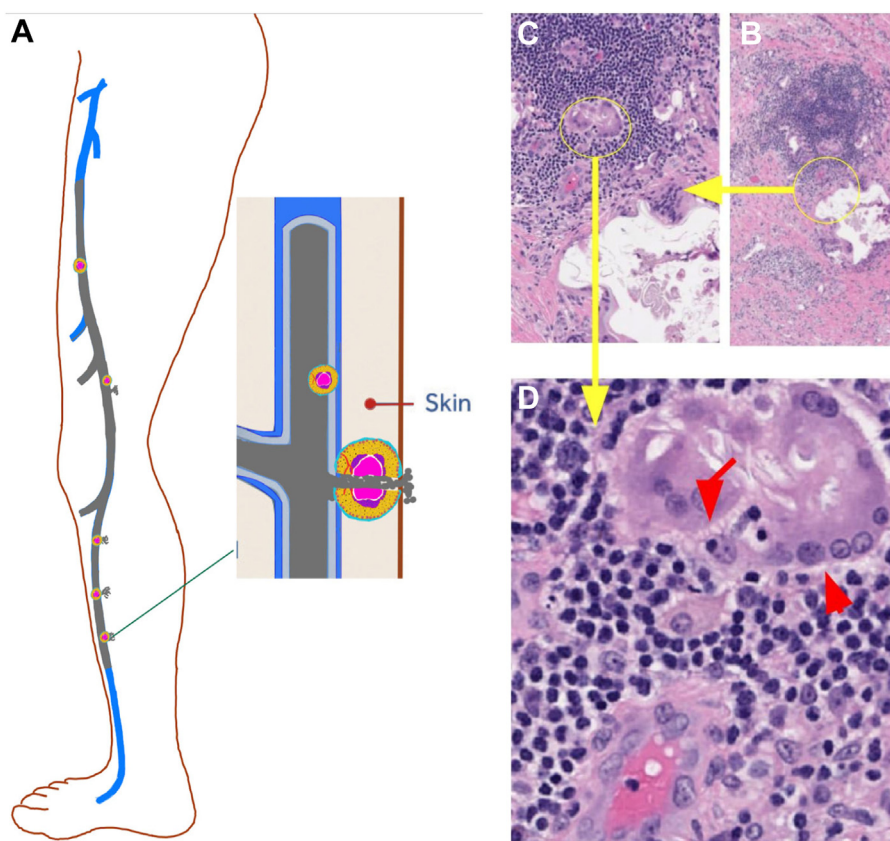


Fig 4. A, Foreign body reaction seen in the vessel lumen, vessel wall, extravascular space, and as a fistula with extrusion of glue. B-D, Histopathologic slides of excised foreign body granuloma at the great saphenous vein (GSV) groin segment. Section shows foreign body giant cells accumulating in areas at different magnifications.

were noted in the tissue surrounding the vessel, and a few were further away in the inflammatory reaction.

Following resection of the groin mass and below-knee GSV, all his skin breakdowns of the treated areas had healed and any erythema and discomfort had resolved. Still, shortly after, he developed a new lesion in the mid- to distal thigh portion with residual VenaSeal in the GSV. Given his past experience, we opted for immediate resection of the troubled area, and he eventually achieved complete healing.

We hypothesize that he developed an exaggerated type IV hypersensitivity reaction and formation of foreign body granulomas. His symptoms improved with immune suppression and recurred after discontinuation of systemic steroids. The groin mass likely represented a well-encapsulated foreign body granuloma. The exaggerated immune response persisted until all foreign material had been removed. Care was taken to avoid excessive intraoperative pressure; however, inadvertent glue extrusion into the subcutaneous space remains a possibility. Alternatively, it is possible that an exaggerated foreign body reaction caused a vein wall defect with the subsequent extrusion of glue.

CASE STUDY 2 (TYPE IV HYPERSENSITIVITY)

A 73-year-old woman presented with pruritic and erythematous skin changes after VenaSeal treatment for CEAP C3 disease at an outside clinic. The patient initially developed a fever with skin erythema and pruritus a few days following her treatment. Around the same time, she developed worsening of her asthma symptoms and opted to start taking prednisone for asthma for ~10 days. Her leg symptoms significantly improved with steroid treatment. However, her symptoms recurred ~1 week later, and she presented for a second opinion. We treated her with a long prednisone taper regimen that lasted for 5 weeks (50 mg, 40 mg, 30 mg, 20 mg, and 10 mg for 1 week each). She had subsequent improvement with this regimen, which resolved her reaction, and she remained symptom-free.

It is unclear whether her symptoms of asthma were provoked by the presence of cyanoacrylate glue or was an independent phenomenon. Regardless, the symptoms of phlebitis responded to systemic steroids that were started for the treatment of asthma and recurred after their discontinuation. A type IV hypersensitivity was suspected, and her signs and symptoms of phlebitis

responded to a prolonged steroid taper regimen. However, we suspect that she remains at risk of episodes of recurrent phlebitis or an exaggerated foreign body reaction in the setting of a secondary proinflammatory state.

RECOMMENDATIONS FOR PATIENT SELECTION AND TREATMENT OF CAC AEs BASED ON THE LITERATURE AND FINDINGS

It might be prudent to avoid the use of CAC for individuals with a history of occupational or other exposure to cyanoacrylate products, autoimmune conditions, atopic dermatitis, recurrent infections, or a history of allergic contact dermatitis to other substances. From our literature review, hypotheses, and case studies, we propose that treatment should be tailored according to the reaction to which the AE is most likely due. A type I hypersensitivity response should be considered in patients with erythema or itching after the procedure. These patients' symptoms can respond to nonsteroidal anti-inflammatory drugs, antihistamines, and topical steroids. However, for individuals with type IV sensitivity or more severe symptoms, as seen in our case 2, a longer steroid taper should be considered. Surgical excision of the entire treated vein could be required to resolve symptoms of a foreign body granuloma.

CONCLUSIONS

We performed a systematic review and proposed mechanisms of how adverse reactions after CAC of the axial saphenous veins can occur. Although CAC, for the most part, is a clinically effective and safe NTNT option, AEs specific to CAC that will require an escalated treatment algorithm can occur, such as the need for prolonged systemic steroid use or vein resection. Thus, clinicians who perform CAC treatment must be aware of these risks, select the appropriate patients for treatment, and be able to recognize and treat AEs should they occur. Further studies are needed to confirm our hypothesis of the proposed mechanisms.

AUTHOR CONTRIBUTIONS

Conception and design: AA, MT, EF

Analysis and interpretation: AA, MT, VS, ML, EF

Data collection: AA, MT, VS, VC, EF

Writing the article: AA, MT, EF

Critical revision of the article: AA, MT, VS, ML, VC, EF

Final approval of the article: AA, MT, VS, ML, VC, EF

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Supplementary Table (online only). Studies reporting foreign body reaction after cyanoacrylate glue closure (CAC)

Study	Symptoms	Product	Duration after CAC, years			
			<1	1	2	>5
Parsi et al, ⁴² 2020	No	VenaBlock	FBGCs	FBGCs	NA	NA
Gibson et al, ³⁸ 2020	Yes	VenaSeal	NA	FBGCs	NA	NA
Almeida et al, ⁴⁶ 2020	No	VenaSeal	NA	NA	NA	FBGC
Langridge et al, ⁴⁵ 2020	Yes	VenaSeal	FBGCs	NA	NA	NA
Park et al, ⁹⁴ 2021	No	VenaSeal	NA	NA	FBGCs	NA

FBGCs, foreign body giant cells; NA, not available.